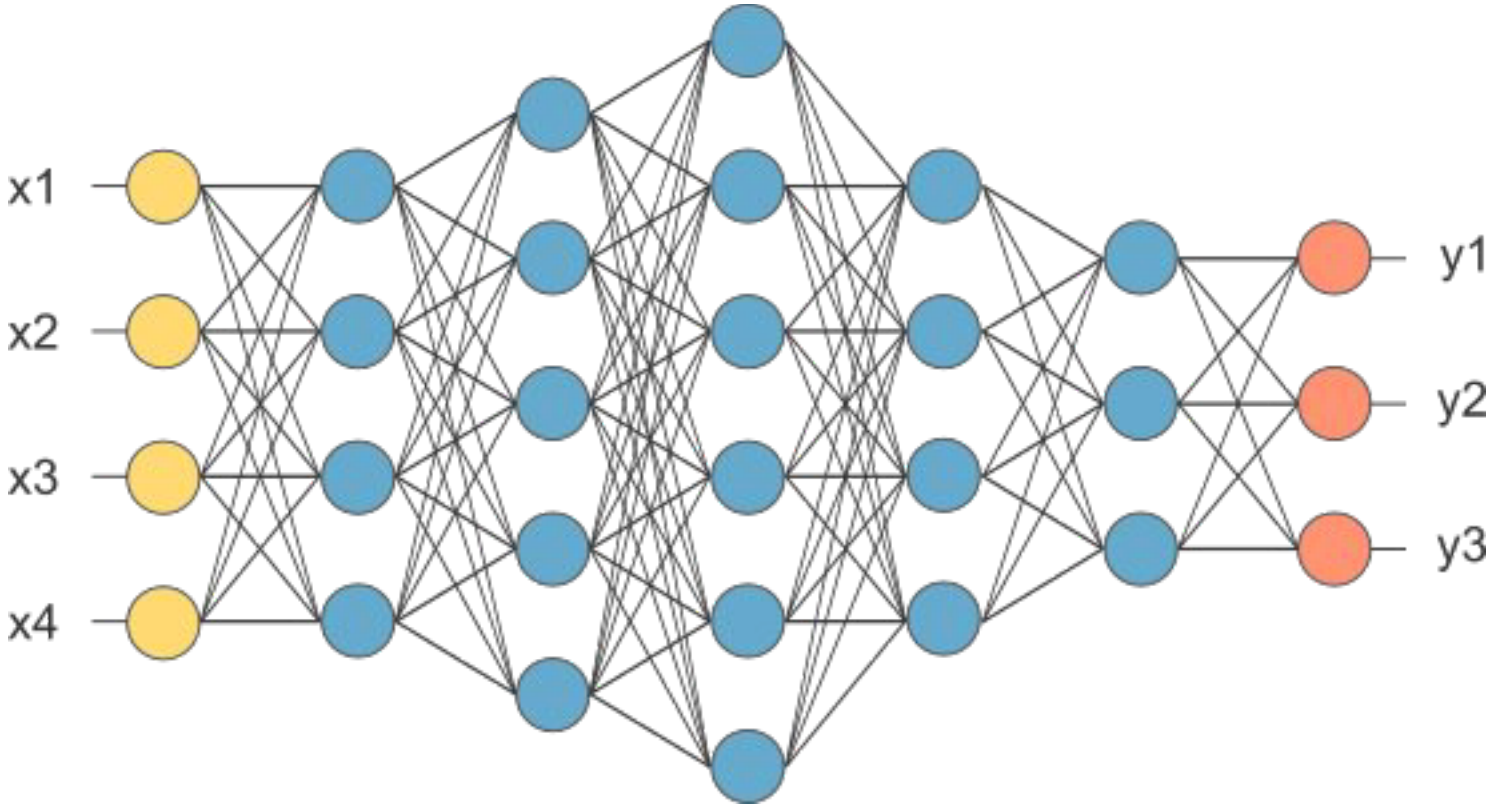
RNN

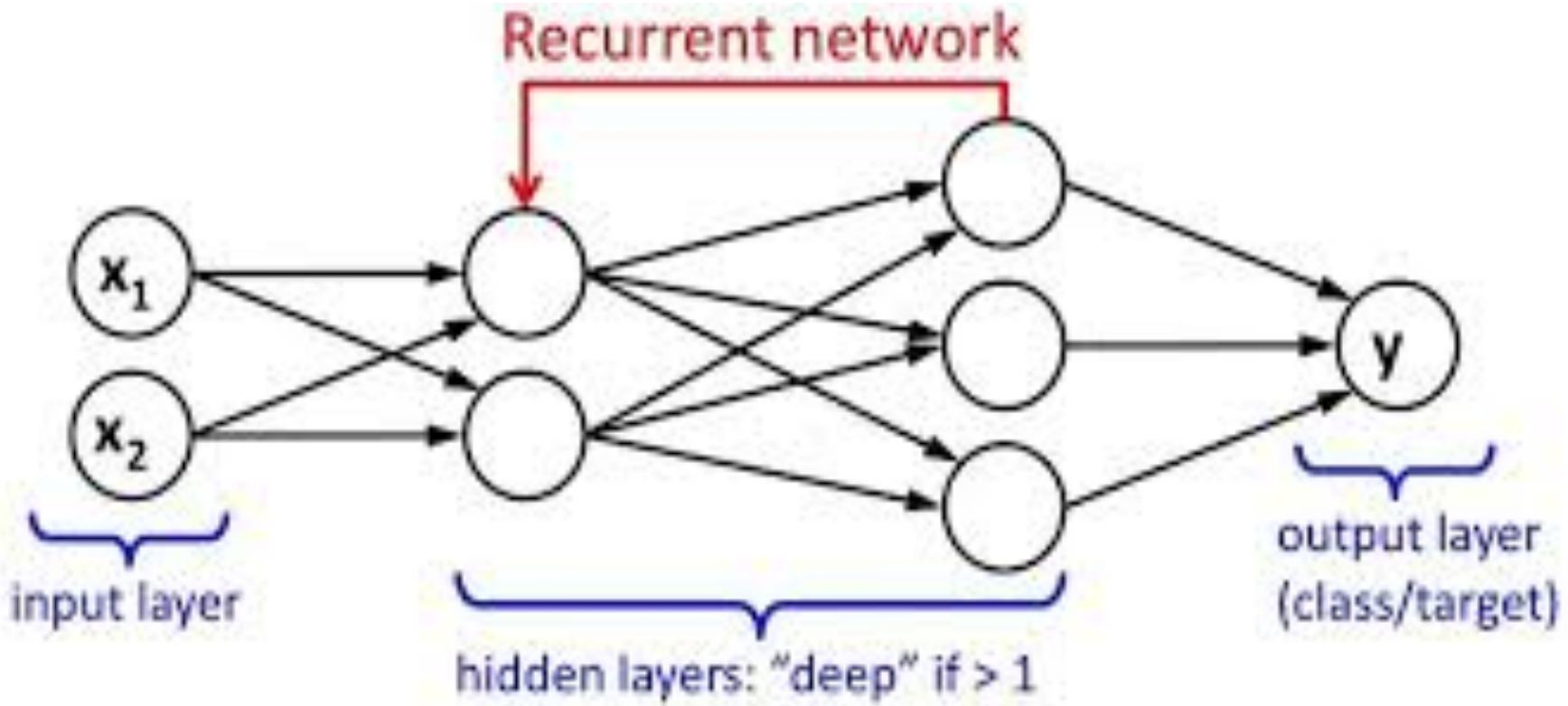


This is our fully connected network. If x_1 ... x_n, n is very large and growing, this network would become too large. We now will input one x_i at a time, and re-use the same edge weights.

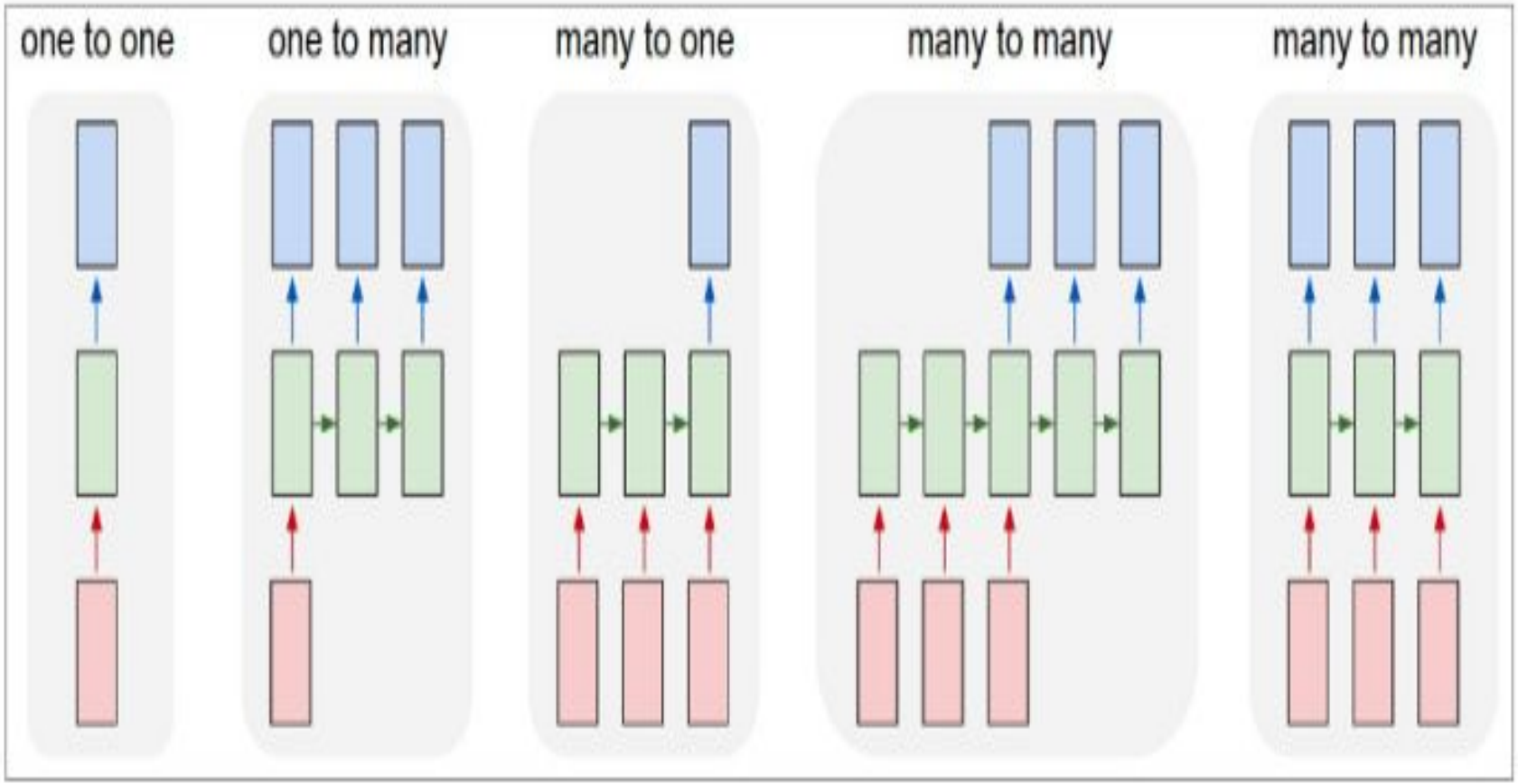
RNN

- A Recurrent Neural Network is a type of neural network that contains loops, allowing information to be stored within the network. In short, Recurrent Neural Networks use their reasoning from previous experiences to inform the upcoming events.

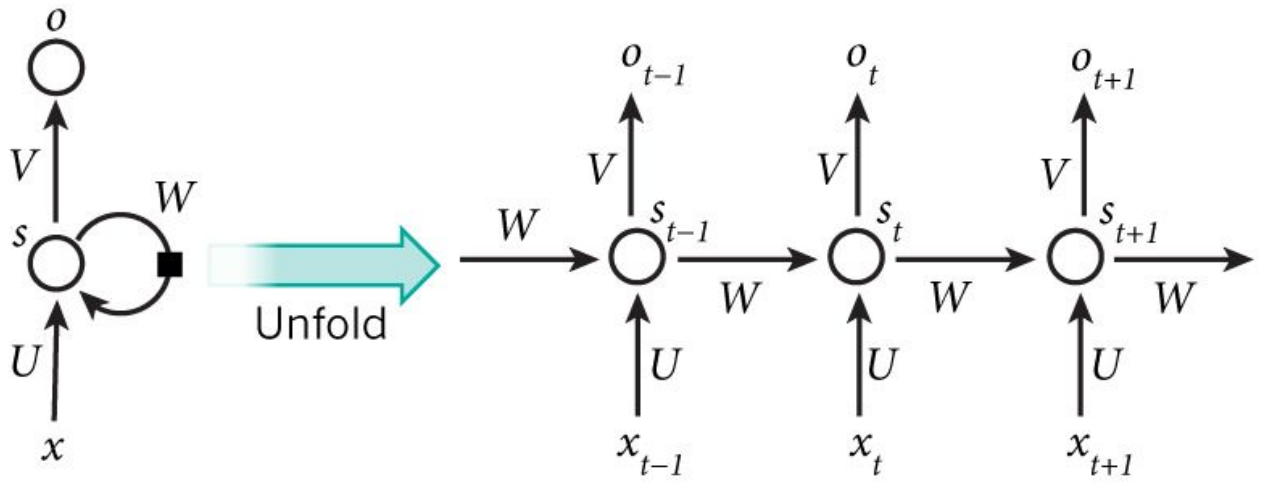
Simple RNN

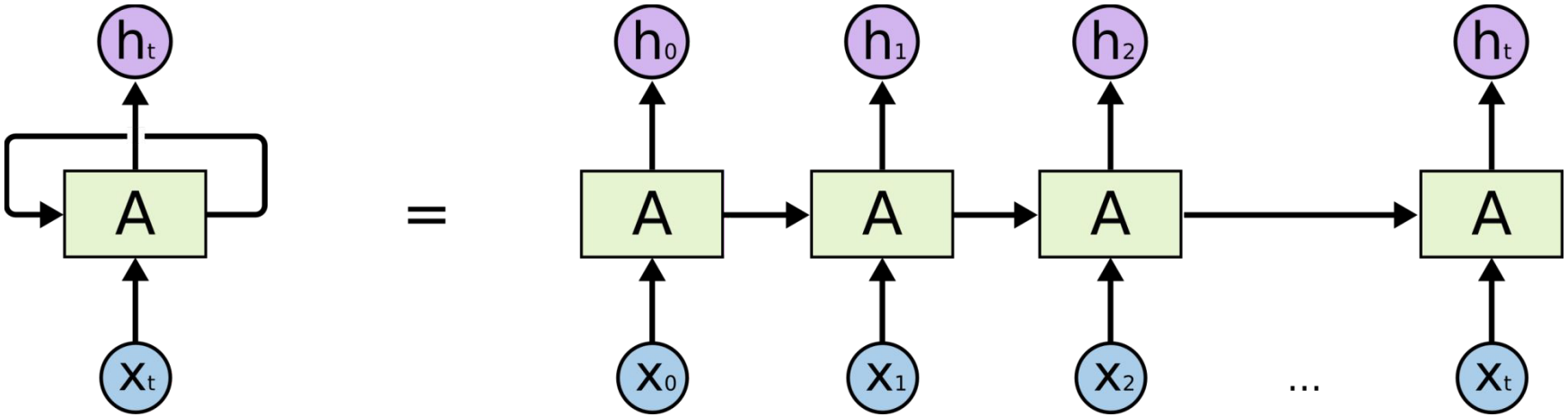

Recurrent network
x_1
x_2
y
input layer
hidden layers: "deep" if > 1
output layer
(class/target)

How do Recurrent Neural Networks work



Recurrent Neural Network

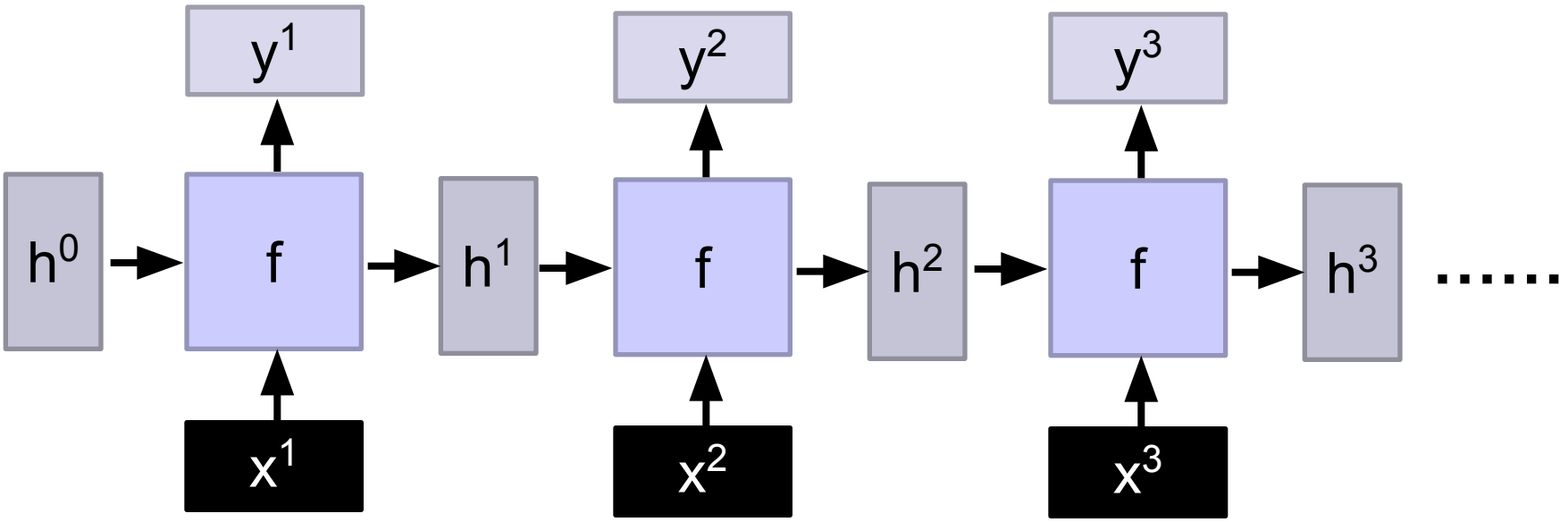




How does RNN reduce complexity?

- Given function f: h',y=f(h,x)

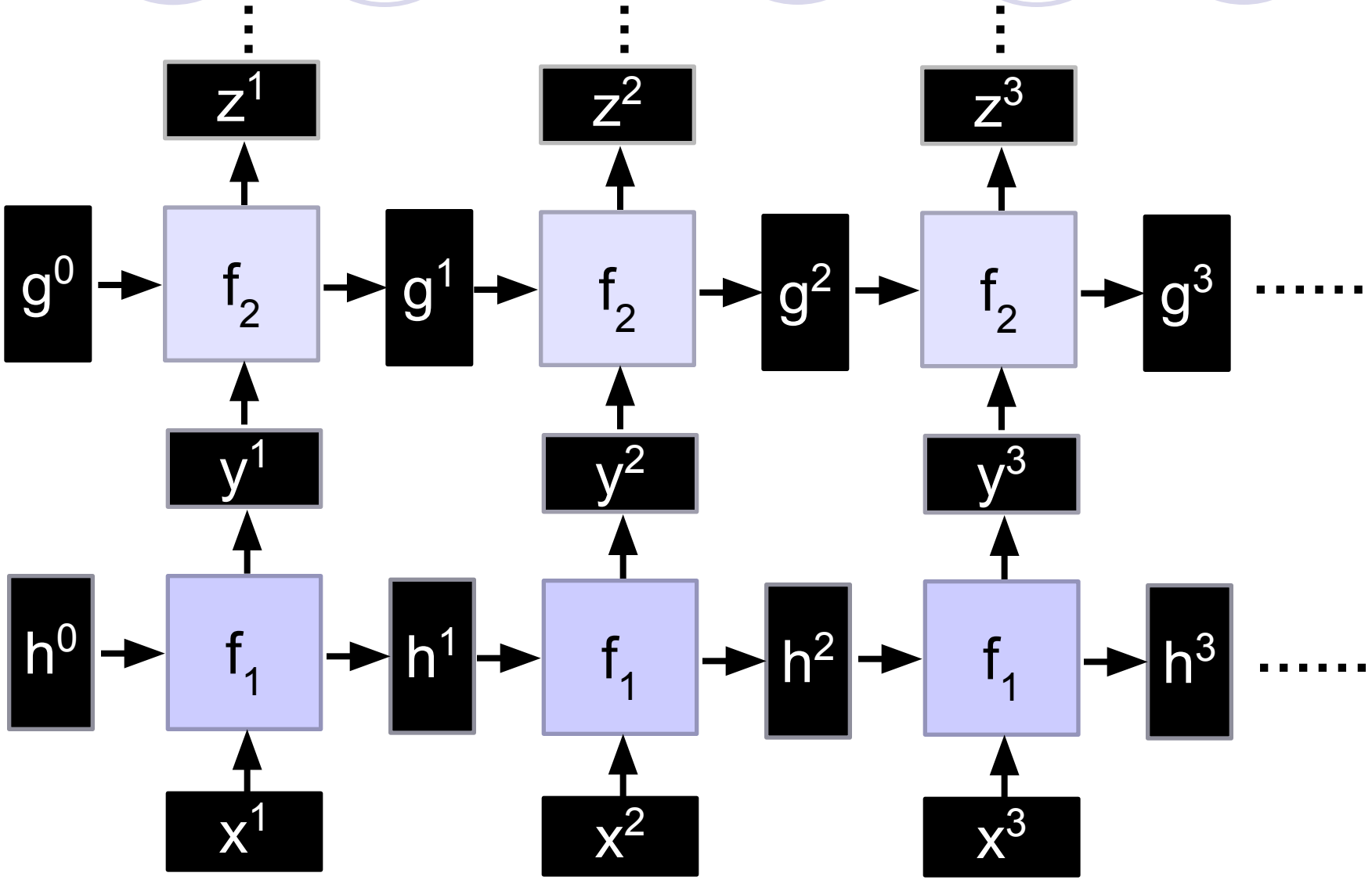
h and h' are vectors with the same dimension



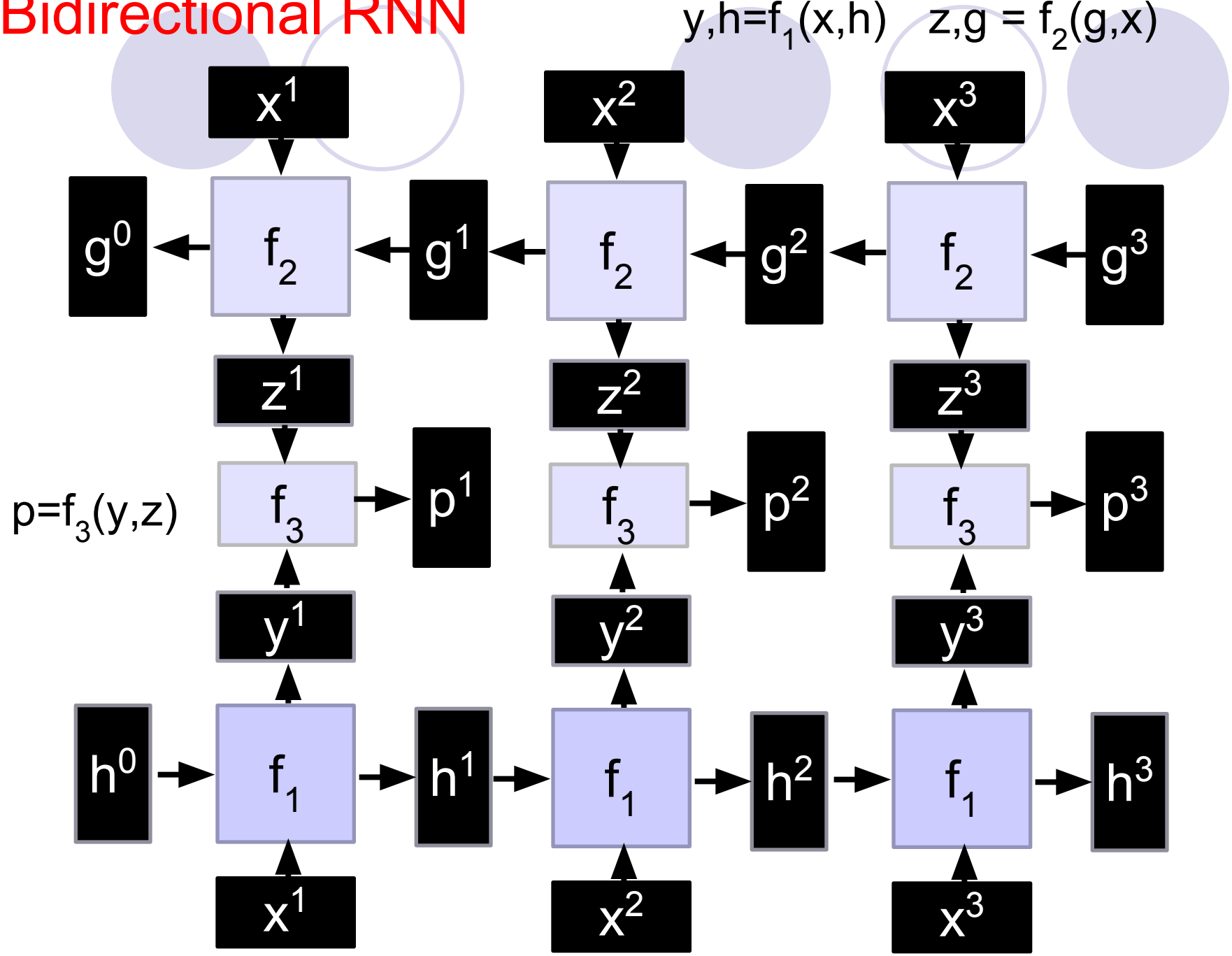
No matter how long the input/output sequence is, we only need one function f. If f's are different, then it becomes a feedforward NN. This may be treated as another compression from fully connected network.

Deep RNN

$h', y = f_1(h, x)$, $g', z = f_2(g, y)$ ...



Bidirectional RNN



Long Term Short Term Memory (LSTM)

- An RNN has short term memory. When used in combination with Long Short Term Memory (LSTM) Gates, the network can have long term memory.
- Instead of the recurring section of an RNN, an LTSM is a small neural network consisting of four neural network layers. These are the recurring layer from the RNN with three networks acting as gates.

LSTM……

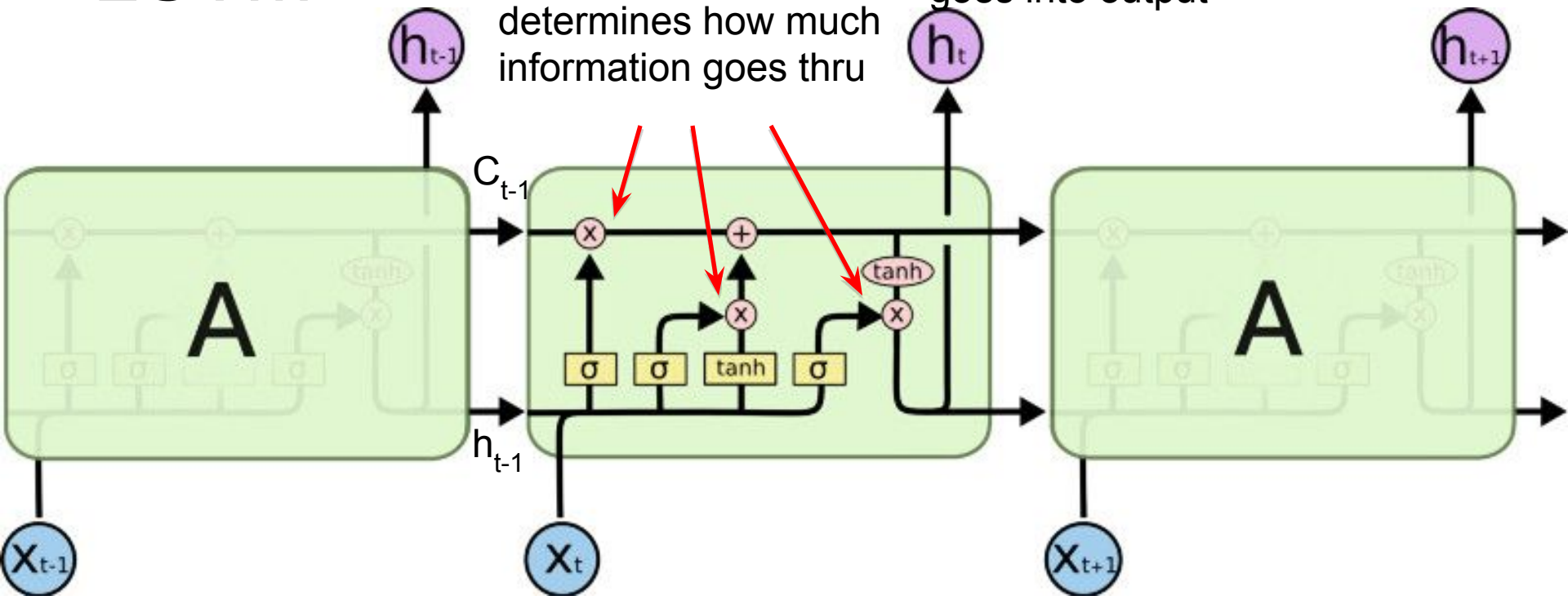
- Long Short Term Memory (LSTM) has three gates.
- An Input gate, this controls the information input at each time step.
- An Output gate, this controls how much information is outputted to the next cell or upward layer
- A Forget gate, this controls how much data to lose at each time step.

LSTM

This decides what info
Is to add to the cell state

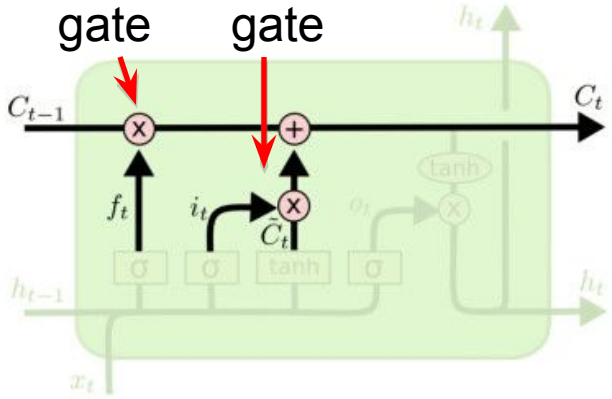
This sigmoid gate
determines how much
information goes thru

Output gate
Controls what
goes into output



Forget gate

input gate



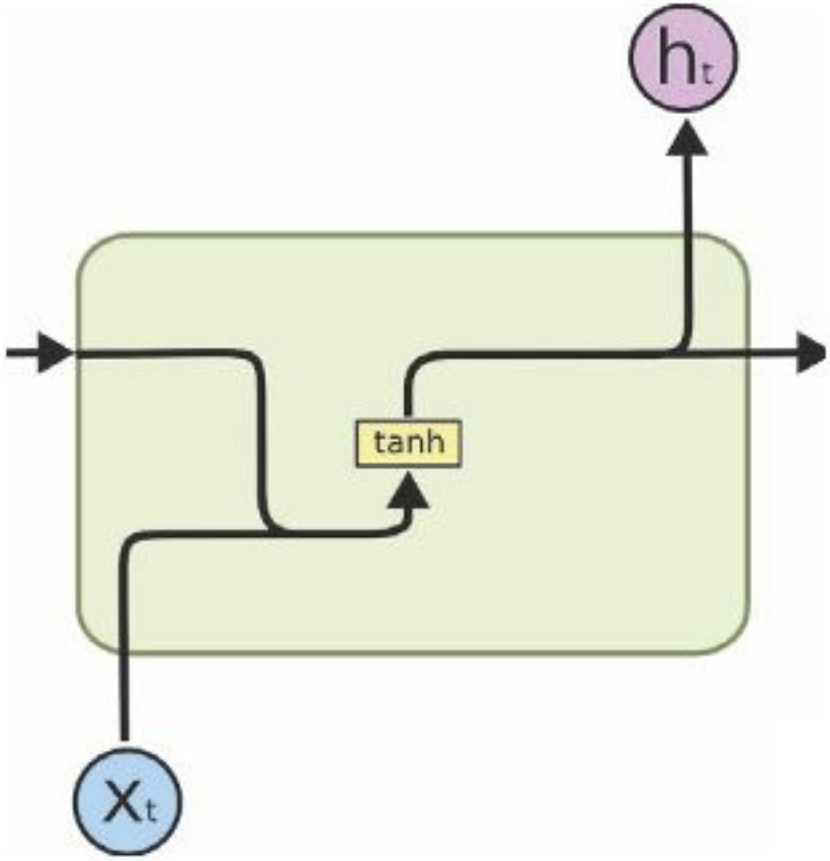
The core idea is this cell state C_t, it is changed slowly, with only minor linear interactions. It is very easy for information to flow along it unchanged.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

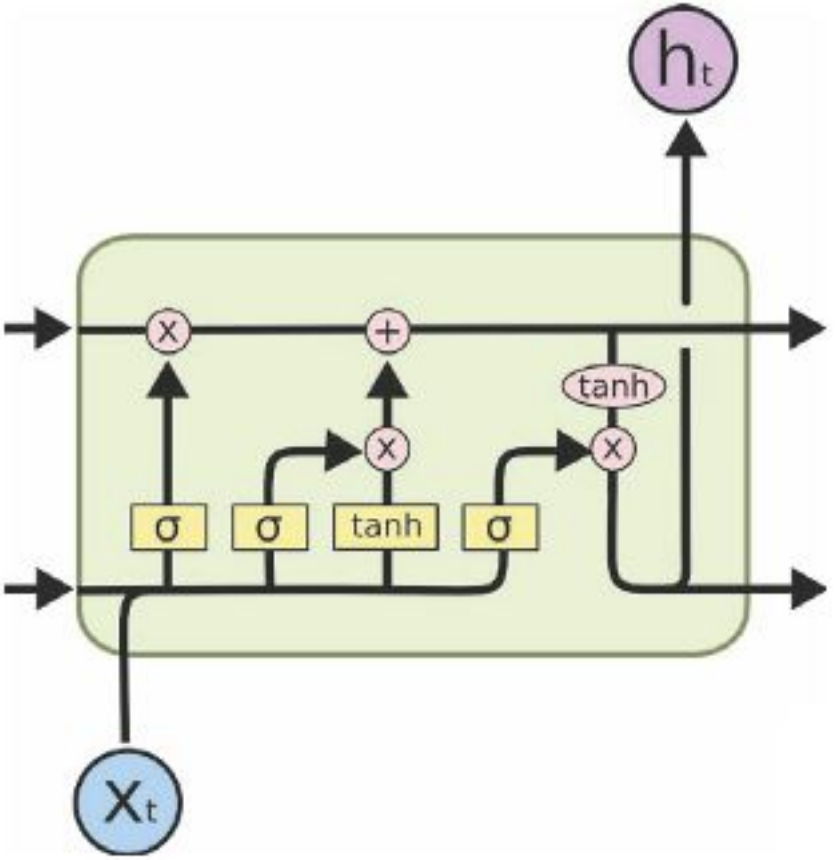
Working of LSTM

- Hidden state and new Input
- Forget gate
- Input gate
- Update cell state
- Update hidden state

RNN vs LSTM



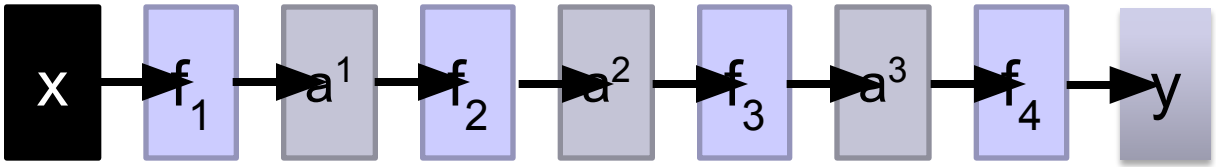
(a) RNN



(b) LSTM

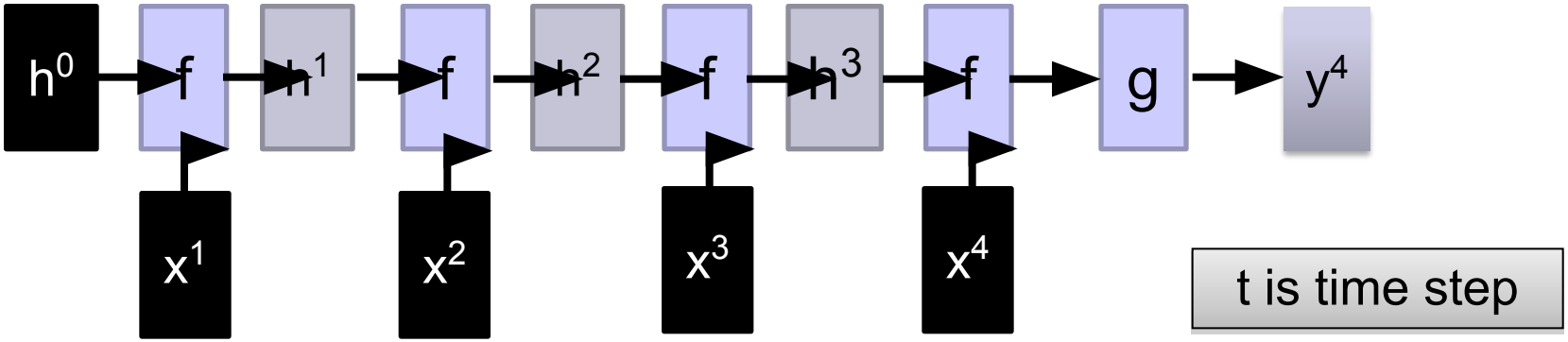
Feed-forward vs Recurrent Network

1. Feedforward network does not have input at each step
2. Feedforward network has different parameters for each layer



t is layer

$a^t = f_t(a^{t-1}) = \sigma(W^t a^{t-1} + b^t)$

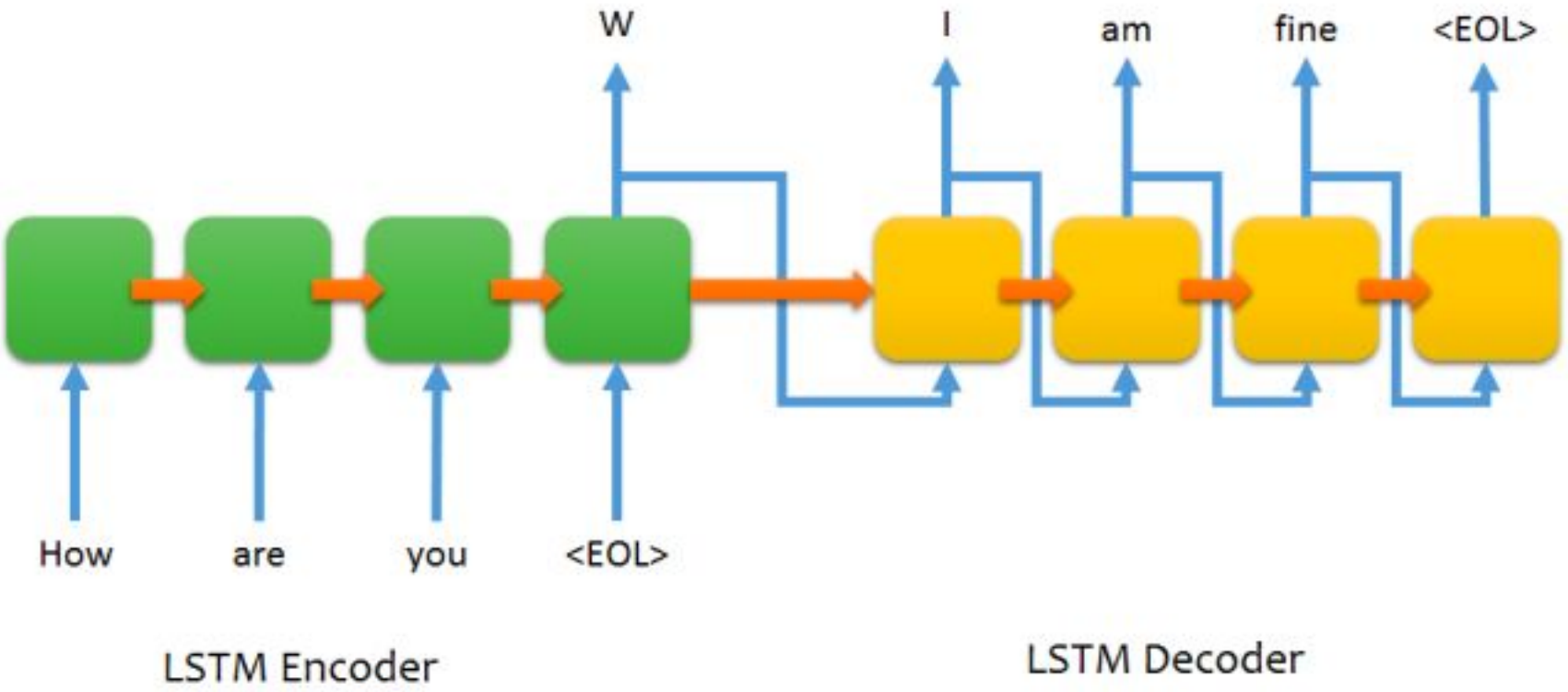


t is time step

$a^t = f(a^{t-1}, x^t) = \sigma(W^h a^{t-1} + W^i x^t + b^i)$

We will turn the recurrent network 90 degrees.

Sequence to sequence chat model

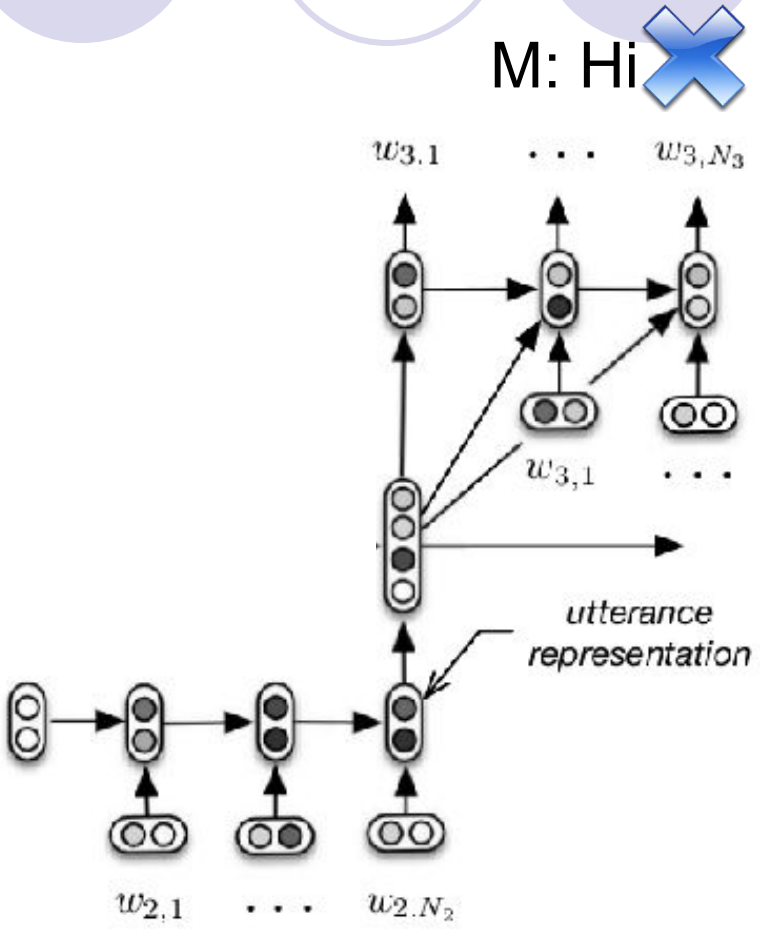


Chat with context

M: Hello

U: Hi

M: Hi



M: Hello

U: Hi

Serban, Iulian V., Alessandro Sordoni, Yoshua Bengio, Aaron Courville, and Joelle Pineau, 2015 "Building End-To-End Dialogue Systems Using Generative Hierarchical Neural Network Models.

Image Caption Generation



A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.



A giraffe standing in a forest with trees in the background.

Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhutdinov, Richard Zemel, Yoshua Bengio, “Show, Attend and Tell: Neural Image Caption Generation with Visual Attention”, ICML, 2015

Image Caption Generation



A large white bird standing in a forest.



A woman holding a clock in her hand.



A man wearing a hat and a hat on a skateboard.



A person is standing on a beach with a surfboard.



A woman is sitting at a table with a large pizza.



A man is talking on his cell phone while another man watches.



Ref: A man and a woman ride a motorcycle

A **man** and a **woman** are **talking** on the **road**



* Possible project?

Ref: A woman is frying food

Someone is **frying** a **fish** in a **pot**